COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212-Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/29/26	**DATE SUBMITTED:**	10/29/26
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Final Hands-on Skills Exam	**Final Hands-on Skills Exam**

1. PROCEDURE OUTPUT

1. Create a repository and label it as "Final_Exam_Surname"

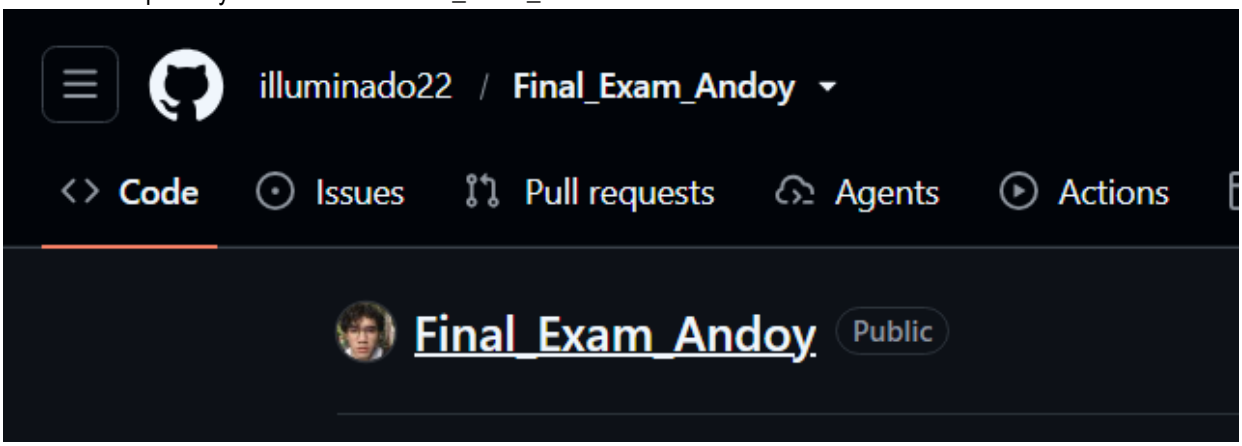


FIgure 1.1: creating repo

2. Clone your new repository in your VM

```
(base) nikko@Nikko:~$ git clone https://github.com/illuminado22/Final_Exam_Andoy.git
Cloning into 'Final_Exam_Andoy'...
warning: You appear to have cloned an empty repository.
(base) nikko@Nikko:~$ cd Final_Exam_Andoy/
(base) nikko@Nikko:~/Final_Exam_Andoy$ ls
(base) nikko@Nikko:~/Final_Exam_Andoy$ git status
On branch main

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

Figure 2.1: checking if git clone worked

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

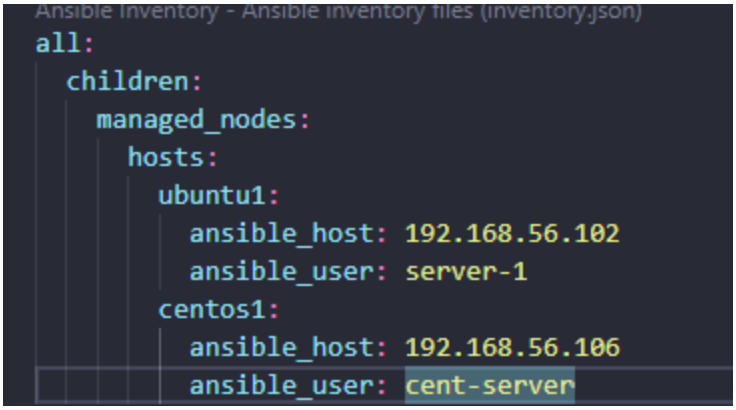


Figure 3: Inventory.yml
Created an inventory file that includes 2 vm which are ubuntu and centOS

```
(base) nikko@Nikko:~/Final_Exam_Andoy$ ansible-inventory -i inventory.yml --list
{
    "_meta": {
        "hostvars": {
            "centos1": {
                "ansible_host": "192.168.56.106",
                "ansible_user": "cent-server"
            },
            "ubuntu1": {
                "ansible_host": "192.168.56.102",
                "ansible_user": "server-1"
            }
        },
        "profile": "inventory_legacy"
    },
    "all": {
        "children": [
            "ungrouped",
            "managed_nodes"
        ]
    },
    "managed_nodes": {
        "hosts": [
            "ubuntu1",
            "centos1"
        ]
    }
}
```

Figure 3.1: list of inventory
Used the ansible-inventory -i inventory.yml –list to list the group that I created in the inventory file and to check if the group was created successfully.

```
(base) nikko@Nikko:~/Final_Exam_Andoy$ ansible all -i inventory.yml -m ping
[WARNING]: Host 'ubuntu1' is using the discovered Python interpreter at '/usr/bin/pyth
s.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for mo
ubuntu1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.12"
    },
    "changed": false,
    "ping": "pong"
}
[WARNING]: Host 'centos1' is using the discovered Python interpreter at '/usr/bin/pyth
.ansible.com/ansible-core/2.19/reference_appendices/interpreter_discovery.html for mor
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": false,
    "ping": "pong"
}
(base) nikko@Nikko:~/Final_Exam_Andoy$
```

Figure 3.2: connectivity of managed nodes

Final_Exam_Andoy > config.yml > web_message

```
1  ansible_managed_user: "Francis Nikko Andoy"
2
3  enterprise_service_name: "nginx"
4  enterprise_port: 80
5  web_message: "Ansible Managed Nginx Enterprise Service"
```

Figure 3.3: config.yml
Contains the services that I will use for this activity.

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

Final_Exam_Andoy > roles > enterprise_service > tasks > main.yml > ...

```yaml
---
Ansible Tasks Schema - Ansible tasks file (ansible.json)
- name: Stop and disable Apache on Ubuntu
  ansible.builtin.service:
    name: apache2
    state: stopped
    enabled: false
  when: ansible_os_family == "Debian"
  ignore_errors: true

- name: Stop and disable httpd on CentOS
  ansible.builtin.service:
    name: httpd
    state: stopped
    enabled: false
  when: ansible_os_family == "RedHat"
  ignore_errors: true

- name: Kill lingering web processes
  ansible.builtin.shell: "fuser -k 80/tcp || true"
  changed_when: false

- name: Install Nginx package
  ansible.builtin.package:
```

Figure 3.1.1: Install nginx service on managed nodes

Since I created a config.yml I just used it like a variables like in the code {{ variableName }}. This is for installing and configuring the nginx

Final_Exam_Andoy > roles > enterprise_service > handlers > main.yml > ...

```yaml
---
Ansible Tasks Schema - Ansible tasks file (ansible.json)
- name: restart_nginx
  ansible.builtin.service:
    name: "{{ enterprise_service_name }}"
    state: restarted
```

FIgure 3.1.2: restart nginx

```
TASK [enterprise_service : Allow enterprise port in firewalld (CentOS)] ********************************************************************
******
skipping: [ubuntu1]
ok: [centos1]

TASK [enterprise_service : Ensure Nginx service is enabled and running] ********************************************************************
******
changed: [ubuntu1]
changed: [centos1]

TASK [Set Message of the Day] ********************************************************************************************************
******
changed: [ubuntu1]
changed: [centos1]

PLAY RECAP ********************************************************************************************************************************
******
centos1                    : ok=9    changed=3    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
ubuntu1                    : ok=9    changed=3    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

Figure 3.1.3: verification of playbook

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

```yaml
roles:

  - role: enterprise_service
    tags: enterprise


  - role: monitoring_service
    tags: monitoring
```

Figure 3.2.1: Added monitoring service in roles

Final_Exam_Andoy > config.yml > node_exporter_version

```yaml
---

ansible_managed_user: "Nikko Andoy"

enterprise_service_name: "nginx"
enterprise_port: 80

monitoring_port: 9100
node_exporter_version: "1.8.2"
```

Figure 3.2.2: modified the config file adding monitoring



Figure 3.2.3: Installation and configuring prometheus

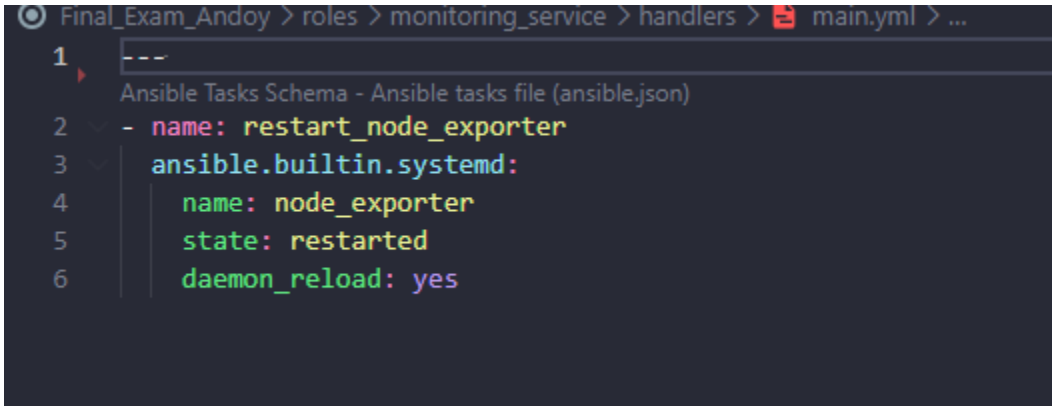


Figure 3.2.4: restarting the prometheus

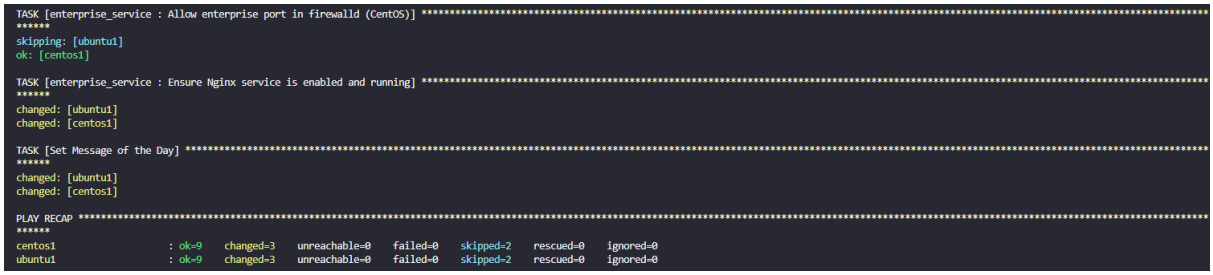


Figure 3.2.5: Verification of installation and config

4.4 Change Motd as "Ansible Managed by <username>"

```yaml
---

ansible_managed_user: "Nikko Andoy"

enterprise_service_name: "nginx"
enterprise_port: 80

monitoring_port: 9100
node_exporter_version: "1.8.2"
```

Figure 4.4.1: changed name

Final_Exam_Andoy > site.yml > { } 0 > [ ] roles

```yaml
---
Ansible Playbook - Ansible playbook files (ansible.json)
- name: Final Exam - Enterprise Service and Monitoring Deployment
  hosts: all
  become: true
  vars_files:
    - config.yml

  tasks:

    - name: Set Message of the Day
      ansible.builtin.copy:
        dest: /etc/motd
        content: "Ansible Managed by {{ ansible_managed_user }}\n"
        mode: '0644'

  roles:

    - role: enterprise_service
      tags: enterprise

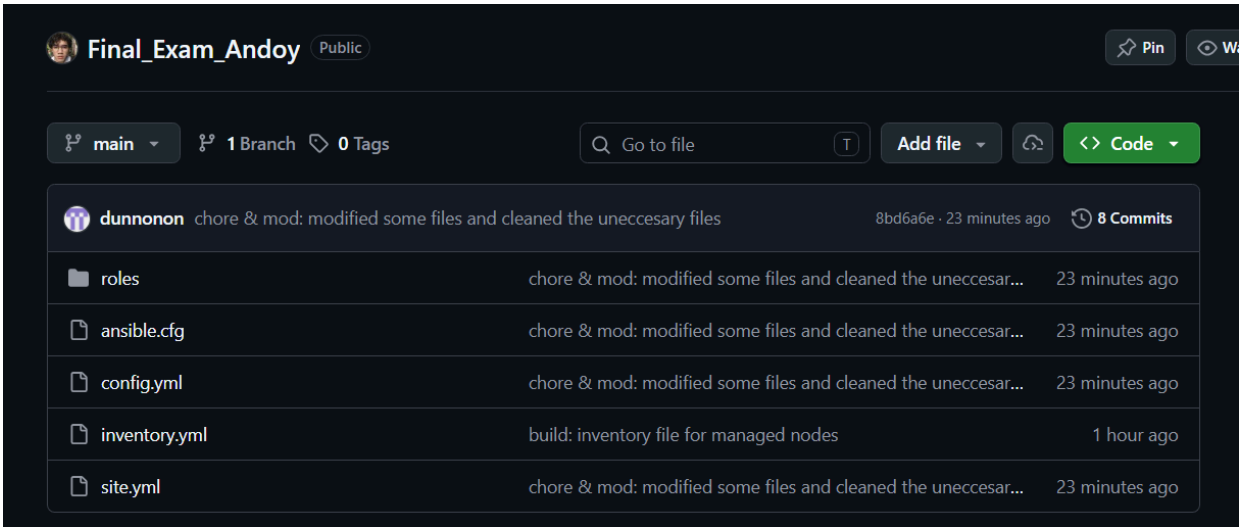
    - role: monitoring_service
      tags: monitoring
```

Figure 4.4.2. site.yml

4. Push and commit your files in GitHub

```
(base) nikko@Nikko:~/Final_Exam_Andoy$ git add roles/monitoring_service/handlers/main.yml
(base) nikko@Nikko:~/Final_Exam_Andoy$ git commit -m "build: restart the prometheus"
[main 8a49490] build: restart the prometheus
 1 file changed, 7 insertions(+)
 create mode 100644 roles/monitoring_service/handlers/main.yml
(base) nikko@Nikko:~/Final_Exam_Andoy$ git push origin main
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 575 bytes | 287.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/illuminado22/Final_Exam_Andoy.git
   8927f2f..8a49490  main -> main
(base) nikko@Nikko:~/Final_Exam_Andoy$ git add roles/monitoring_service/tasks/main.yml
(base) nikko@Nikko:~/Final_Exam_Andoy$ git commit -m "build: install and configure prometheus"
[main a6ebb6c] build: install and configure prometheus
 1 file changed, 77 insertions(+)
 create mode 100644 roles/monitoring_service/tasks/main.yml
(base) nikko@Nikko:~/Final_Exam_Andoy$ git push origin main
Enumerating objects: 9, done.
```

```
(base) nikko@Nikko:~/Final_Exam_Andoy$ git commit -m "chore & mod: modified some files and cleaned the uneccesary files"
[main 8bd6a6e] chore & mod: modified some files and cleaned the uneccesary files
 5 files changed, 30 insertions(+), 13 deletions(-)
 create mode 100644 site.yml
(base) nikko@Nikko:~/Final_Exam_Andoy$ git push origin main
Enumerating objects: 20, done.
Counting objects: 100% (20/20), done.
Delta compression using up to 12 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (11/11), 1.17 KiB | 598.00 KiB/s, done.
Total 11 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 3 local objects.
To https://github.com/illuminado22/Final_Exam_Andoy.git
   a6ebb6c..8bd6a6e  main -> main
(base) nikko@Nikko:~/Final_Exam_Andoy$
```



5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

```
TASK [enterprise_service : Allow enterprise port in firewalld (CentOS)] ********************************************************************************
******
skipping: [ubuntu1]
ok: [centos1]

TASK [enterprise_service : Ensure Nginx service is enabled and running] ********************************************************************************
******
changed: [ubuntu1]
changed: [centos1]

TASK [Set Message of the Day] ********************************************************************************************************************
******
changed: [ubuntu1]
changed: [centos1]

PLAY RECAP ********************************************************************************************************************
******
centos1                    : ok=9    changed=3    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
ubuntu1                    : ok=9    changed=3    unreachable=0    failed=0    skipped=2    rescued=0    ignored=0
```

Figure 5.1. Enterprise service

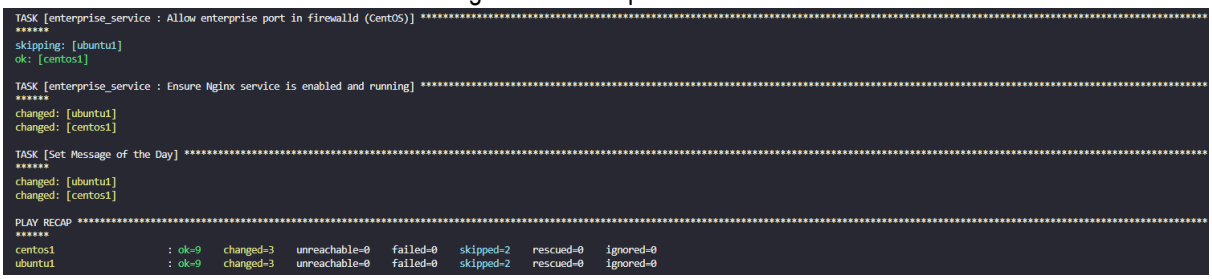


Figure 5.2. Monitoring

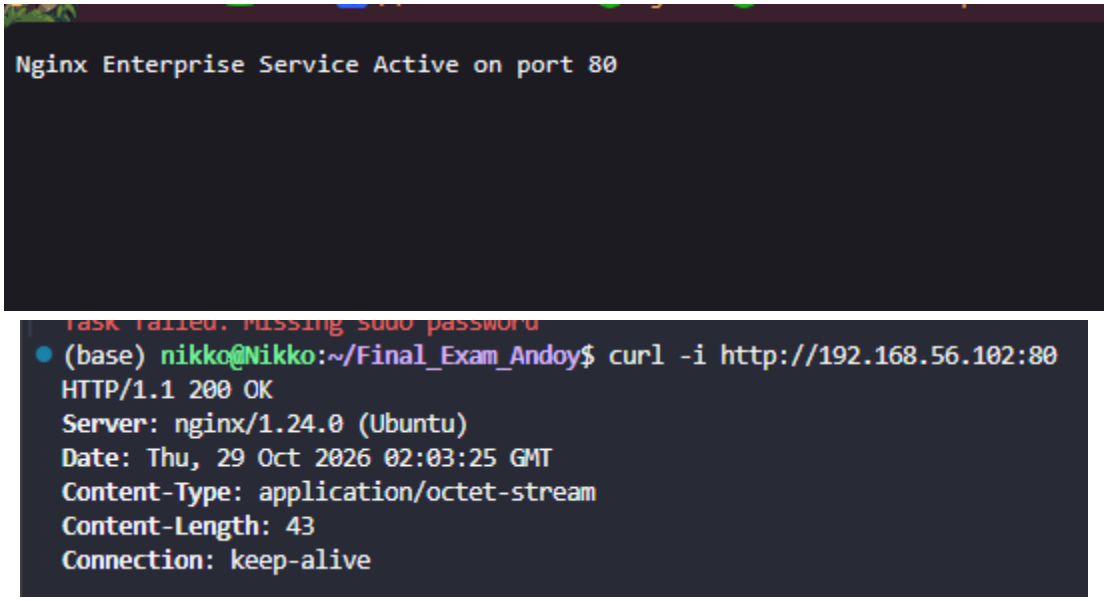
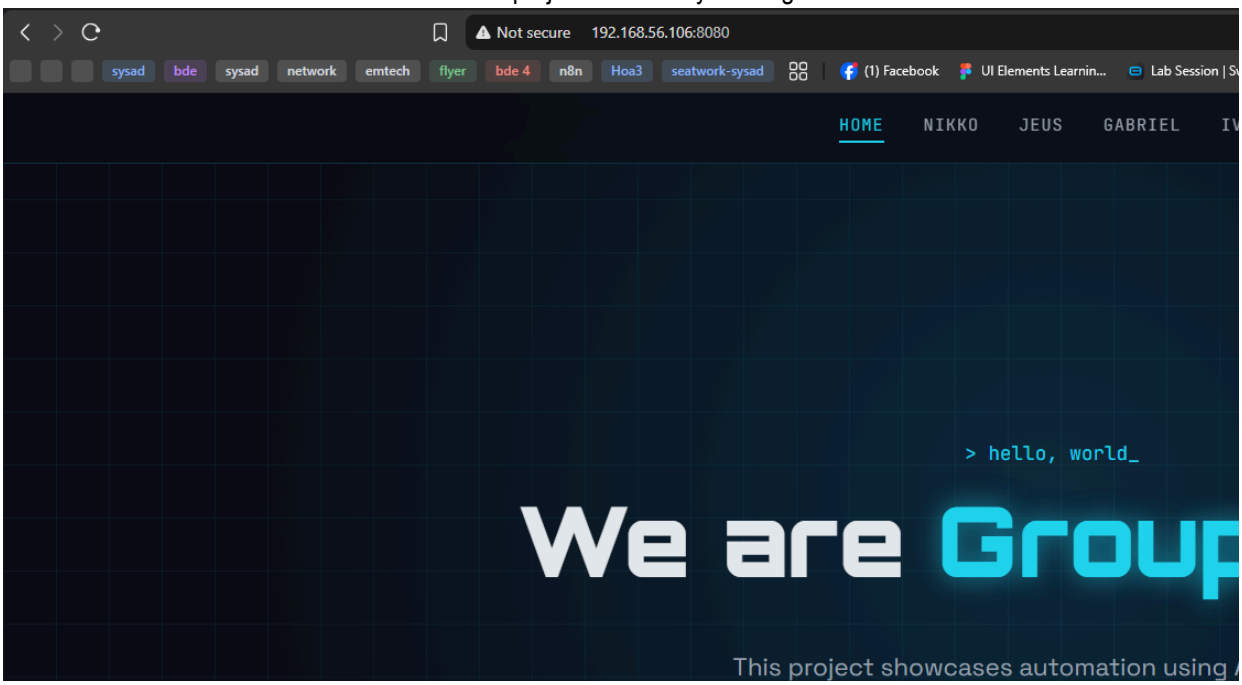


Figure 5.3. Verification of nginx (ubuntu)

```
(base) nikko@Nikko:~/Final_Exam_Andoy$ curl -i http://192.168.56.106:8080
HTTP/1.1 200 OK
Server: nginx/1.31.6
Date: Thu, 29 Oct 2026 02:10:41 GMT
Content-Type: text/html
Content-Length: 700
Last-Modified: Tue, 06 Oct 2026 10:19:48 GMT
Connection: keep-alive
ETag: "6ac4cb44-2bc"
Accept-Ranges: bytes
```

Figure 5.4. Verification of prometheus (centOS)

I used curl because our final project is currently running on the centOS server



5. For your final exam to be counted, please paste your repository link as an answer in this exam.

https://github.com/illuminado22/Final_Exam_Andoy

Conclusion:
In conclusion I successfully configured and installed the nginx and prometheus for enterprise services and monitoring across ubuntu and centOS. The installation and configuration is automated using yaml and were confirmed using curl. And lastly the pushing the files into my github repo

Note: Extra points if you will implement the said services via containerization.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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